Cross Validation

Experiment 1:
Using the Titanic dataset, split the data into 80% training and 20% testing sets. Train a Random Forest classifier and evaluate its accuracy.

Questions:

1. What are the sizes of the training and testing datasets?
2. Calculate the classification accuracy.
3. How does changing the test size from 20% to 30% affect the result?

Experiment 2:
Implement 5-Fold Cross Validation using the Titanic dataset and a Random Forest classifier.

Questions:

1. How many samples are used for training and testing in each fold?
2. Compute the accuracy for each fold.
3. Calculate the average accuracy across all folds.
4. Compare the result with Hold-Out Validation.

Experiment 3:
Use Stratified 5-Fold Cross Validation on the Titanic dataset with a Random Forest classifier.

Questions:

1. Calculate the average classification accuracy.
2. Compare the results with ordinary K-Fold Cross Validation.

Experiment 4:
Implement Leave-One-Out Cross Validation using the Titanic dataset and a Random Forest classifier.

Questions:

1. How many iterations are performed?
2. What is the computational cost compared to 5-Fold CV?
3. Calculate the overall accuracy.

Experiment 5:
Apply Repeated 10-Fold Cross Validation (3 repeats) using the Titanic dataset and a Random Forest classifier.

Questions:

1. How many total training/testing cycles are performed?
2. Calculate the Mean Squared Error (MSE) for each repetition.
3. Determine the average MSE.